This assignment will not be graded and is only for practice.

1) Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B. ***1 point***

	Functions of two variables		Direction of maximum directional derivative at (0,0)
(1)	$f(x,y) = x^2 - xy + y^2$	(i)	(0,-1)
(2)	$f(x,y) = 3x^2 + 3x + y$	(ii)	(1,-3)
(3)	$f(x,y) = e^{x+y} - 4y$	(iii)	(0,0)
(4)	$f(x,y) = sin(y) + cos(x)$	(iv)	(3,1)
		(v)	(0,1)
		(vi)	(1,3)

Table: M2PA10.1

- [ ] $(1) \longrightarrow (v)$
- [ ] $(1) \longrightarrow (iii)$
- [ ] $(2) \longrightarrow (iv)$
- [ ] $(3) \longrightarrow (iv)$
- [ ] $(2) \longrightarrow (vi)$
- [ ] $(4) \longrightarrow (v)$
- [ ] $(3) \longrightarrow (vi)$
- [ ] $(4) \longrightarrow (i)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1) \longrightarrow (iii)$
$(2) \longrightarrow (iv)$
$(4) \longrightarrow (v)$

2) Which of the following statements is(are) correct? **1 point**

- [ ] Maximum directional derivative at (0,0) for the function $f(x,y) = |x| + |y|$ is 1.
- [ ] The graph of the linear approximation to a function $f(x,y)$ at a given point P is the tangent plane to $f(x,y)$ at the point P.
- [ ] A tangent plane to a function $f(x,y)$ exists at every point.
- [ ] All directional derivatives of a function $f(x,y)$ at a point P are 0 if the tangent plane is parallel to the XY-plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The graph of the linear approximation to a function $f(x,y)$ at a given point P is the tangent plane to $f(x,y)$ at the point P.
All directional derivatives of a function $f(x,y)$ at a point P are 0 if the tangent plane is parallel to the XY-plane.

3) Which of the following is(are) critical points of the scalar valued multivariable function $f(x,y) = 6x^2 - 2x^3 + 3y^2 - 6xy$? **1 point**

- [ ] (1,1)
- [ ] (0,0)
- [ ] (1,-1)
- [ ] No critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1,1)
(0,0)

4) Consider two functions f_1, f_2 defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).$$

If $L_{f_1}(x, y, z)$ and $L_{f_2}(x, y, z)$ are the linear approximation of $f_1(x, y, z)$ and $f_2(x, y, z)$ at point (1, -1, 1), then find the value of $85\,(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z))$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 30.0

1 point

5) $z = f(x, y) = \frac{-7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}$ represents a surface. Let (x_0, y_0, z_0) be the point where the tangent plane to the given surface is parallel to the plane $7x - 8y + 3z + 2 = 0$. Find the value of $11x_0 + 5y_0 + 4z_0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 55

1 point

6) Find the number of points where the tangent plane to the function $f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$ at the points is parallel to the XY-plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Suppose $ax + by + cz = d_1$ is a tangent plane to the function $f(x, y) = x^3 + 3y^3 + 9xy$ at (-1,2) and $ax + by + cz = d_2$ is a tangent plane to the same function $f(x, y)$ at $(\alpha, 1)$.

Use the above information to answer questions 7-8:

7) Which of the following options is(are) true? ***1 point***

- [ ] $\alpha = -2$
- [ ] $\alpha = 2$
- [ ] There are only two possible candidates for α.
- [ ] $|d_2 - d_1| = 12$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\alpha = 2$

$|d_2 - d_1| = 12$

8) Find the value of $\frac{2a+b+c}{d_1+d_2}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Let $L(x, y)$ be the linear approximation to $f(x, y) = xe^y + ln(x + y)$ at point (1,0).

Use the above information to answer questions 9-10:

9) Suppose $L(x, y) = ax + by + c$, where $a, b, c \in \mathbb{R}$. Find the value of $a + b - c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

10) Find the approximate value of the function $f(x, y)$ at (1.1,0.1) using the linear approximation function $L(x, y)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1.4

1 point

Your score is: 0/10